Numerical Solver Parity & CSI Commercial Verification

This table documents the quantitative parity verification between the custom deterministic symbolic engines (`fem3d`, `structdyn`) and commercial finite element software (CSI SAP2000 v24 and CSI ETABS v21). All discrepancies are well below the target peer-review acceptance threshold of $< 0.50\%$.

Structural Model	Verification Quantity	Unit	Commercial Reference (CSI)	Symbolic Engine (`fem3d`/`structdyn`)	Discrepancy (%)	Parity Status
Portal Frame (SAP2000)	Total Base Shear / Vertical Reaction Fz	kN	140.000	140.000	0.000%	PASSED (< 0.5%)
Portal Frame (SAP2000)	Column Base Shear Fx (Joint 1)	kN	12.928	12.928	0.001%	PASSED (< 0.5%)
Portal Frame (SAP2000)	Column Base Moment My (Joint 1)	kN*m	12.826	12.826	0.002%	PASSED (< 0.5%)
Portal Frame (SAP2000)	Joint 3 Vertical Deflection Uz	mm	0.0525	0.0525	0.000%	PASSED (< 0.5%)
Portal Frame (SAP2000)	Joint 3 Lateral Sway Deflection Ux	mm	0.0086	0.0086	0.217%	PASSED (< 0.5%)
Portal Frame (SAP2000)	Fundamental Period T1 (Mode 1)	s	0.0931	0.0931	0.036%	PASSED (< 0.5%)
Portal Frame (SAP2000)	Fundamental Period T2 (Mode 2)	s	0.0558	0.0558	0.015%	PASSED (< 0.5%)
Portal Frame (SAP2000)	Fundamental Period T3 (Mode 3)	s	0.0504	0.0504	0.026%	PASSED (< 0.5%)
Portal Frame (ETABS)	Total Vertical Base Reaction Fz	kN	60.000	60.000	0.000%	PASSED (< 0.5%)
Portal Frame (ETABS)	Column Axial Reaction (Joint 1)	kN	30.000	30.000	0.000%	PASSED (< 0.5%)

Structural Model	Verification Quantity	Unit	Commercial Reference (CSI)	Symbolic Engine (`fem3d`/`structdyn`)	Discrepancy (%)	Parity Status
Portal Frame (ETABS)	Joint 3 Axial Column Shortening Uz	mm	0.0173	0.0173	0.093%	PASSED (< 0.5%)
3D 3-Story Frame (SAP2000)	Total DEAD Vertical Base Reaction Fz	kip	809.338	809.338	0.000%	PASSED (< 0.5%)
3D 3-Story Frame (SAP2000)	Total LIVE Vertical Base Reaction Fz	kip	670.965	670.965	0.000%	PASSED (< 0.5%)
3D 3-Story Frame (SAP2000)	Max DEAD Vertical Deflection Uz	in	0.0169	0.0169	0.177%	PASSED (< 0.5%)
3D 3-Story Frame (SAP2000)	Max LIVE Vertical Deflection Uz	in	0.0150	0.0150	0.068%	PASSED (< 0.5%)
3D 3-Story Frame (SAP2000)	Fundamental Period T1 (Mode 1)	s	0.4337	0.4337	0.009%	PASSED (< 0.5%)
3D 3-Story Frame (SAP2000)	Fundamental Period T2 (Mode 2)	s	0.4164	0.4164	0.002%	PASSED (< 0.5%)
3D 3-Story Frame (SAP2000)	Fundamental Period T3 (Mode 3)	s	0.3961	0.3961	0.012%	PASSED (< 0.5%)

Formatted LaTeX Table (`table_solver_verification.tex`)

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\begin{table*}[t]
\centering
\caption{Numerical Parity Verification Between Custom Symbolic Solvers
(\texttt{fem3d}, \texttt{structdyn}) and Commercial Finite Element Suites (CSI
SAP2000 and ETABS).}
\label{tab:solver_verification}
\small
\begin{tabular}{llcccc}
\hline
\textbf{Benchmark Structure} & \textbf{Physical / Dynamic Parameter} &
\textbf{Unit} & \textbf{CSI Reference} & \textbf{Symbolic Solvers} &
\textbf{Discrepancy (\%)} \\
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\hline
Portal Frame (SAP2000) & Total Base Shear / Vertical Reaction Fz & kN & 140.000 &
140.000 & 0.000\% \\
 & Column Base Shear Fx (Joint 1) & kN & 12.928 & 12.928 & 0.001\% \\
 & Column Base Moment My (Joint 1) & kN*m & 12.826 & 12.826 & 0.002\% \\
 & Joint 3 Vertical Deflection Uz & mm & 0.0525 & 0.0525 & 0.000\% \\
 & Joint 3 Lateral Sway Deflection Ux & mm & 0.0086 & 0.0086 & 0.217\% \\
 & Fundamental Period T1 (Mode 1) & s & 0.0931 & 0.0931 & 0.036\% \\
 & Fundamental Period T2 (Mode 2) & s & 0.0558 & 0.0558 & 0.015\% \\
 & Fundamental Period T3 (Mode 3) & s & 0.0504 & 0.0504 & 0.026\% \\
Portal Frame (ETABS) & Total Vertical Base Reaction Fz & kN & 60.000 & 60.000 &
0.000\% \\
 & Column Axial Reaction (Joint 1) & kN & 30.000 & 30.000 & 0.000\% \\
 & Joint 3 Axial Column Shortening Uz & mm & 0.0173 & 0.0173 & 0.093\% \\
3D 3-Story Frame (SAP2000) & Total DEAD Vertical Base Reaction Fz & kip & 809.338
& 809.338 & 0.000\% \\
 & Total LIVE Vertical Base Reaction Fz & kip & 670.965 & 670.965 & 0.000\% \\
 & Max DEAD Vertical Deflection Uz & in & 0.0169 & 0.0169 & 0.177\% \\
 & Max LIVE Vertical Deflection Uz & in & 0.0150 & 0.0150 & 0.068\% \\
 & Fundamental Period T1 (Mode 1) & s & 0.4337 & 0.4337 & 0.009\% \\
 & Fundamental Period T2 (Mode 2) & s & 0.4164 & 0.4164 & 0.002\% \\
 & Fundamental Period T3 (Mode 3) & s & 0.3961 & 0.3961 & 0.012\% \\
\hline
\end{tabular}
\end{table*}
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